Algorithm 1 Restaurant Scheduler: Build One Day's Schedule (Version 0)

Input: employee file; staffing-needs file
Output: a valid schedule with breaks, or the role and time that cannot be covered
Read the files — Alberto, Ismael
read the employee file
if a required field is missing **then**
    print "input error"; **stop**
end if
read the staffing-needs file
if a time block or count is invalid **then**
    print "input error"; **stop**
end if
Build the schedule — Pablo
for each 30-minute time block, in order **do**
    for each role needed in that block **do**
        list employees who have the role, are available the whole block, are not already assigned, and stay within max hours
        if the list is not empty **then**
            pick the eligible employee with the fewest roles; if tied, pick the first person on the list
            assign them to the role in that block
        else
            go back to the most recent employee choice and try the next eligible employee
            stop when all choices have been tried **and mark this role and block "cannot be covered"**
        end if
    end for
end for
for each employee whose shift is longer than 6 hours **do** ▷ **a shift = one employee's back-to-back blocks**
    place a 30-minute break 3 hours after the employee's shift starts
    if the break is not covered **then**
        choose another available employee for that role
        if none is available, move the break to the next 30-minute block, **until the end of the shift**
        if no block works, mark the break "cannot be covered"
    end if
end for
Check the schedule — Samina
check availability, roles, coverage, max hours, overlaps, and breaks
if a rule fails **then**
    go to the time block with the problem and try another eligible employee
    if none works, go back to **the most recent** earlier employee choice
end if
Print the result — Naveed
if anything is "cannot be covered" **then**
    print "No valid schedule." and each role and time not covered, **joining back-to-back blocks (12:00–12:30 and 12:30–1:00 print as 12–1)**
else
    print the valid schedule: **names, roles, shift times, and breaks**
end if

Red: added or changed from your submission. Everything else is your own work.